

Programme	B. Sc. STATISTICS				
Course Code	STA 8 CJ 489				
Course Title	RESEARCH METHODOLOGY				
Type of Course	Major				
Semester	VIII				
Academic Level	400-499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	Basic knowledge of typesetting & publishing				
Course Summary	To understand the concept of Research, presentation & Publication.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
1.	Identify the Concept of Research in Statistics, Selection of Topics, Perform Literature Review	U	F	Seminar Presentation /
2.	Express Scientific Word Processing with LaTeX and MS-Word	U	F	Instructor-created exams
3.	Develop Simulation.	Ap	C	Group Tutorial Work
4.	Compute Computer Oriented Numerical Methods	Ap	C	Home Assignments
5.	Describe Plagiarism	U	F	Practical Assignment
6.	Write Thesis	Ap	P	Observation of Practical Skills
* - Remember (R), Understand (U), Apply (Ap), Analyze (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (60)	Marks (70)
I	Introduction to Research Methodology in Statistics		15	20
	1	Meaning of Research, Objectives of Research		
	2	Types of Research- Descriptive Vs. Analytical, Applied Vs Fundamental, Quantitative Vs Qualitative, Conceptual Vs Empirical		
	3	Concept of Research in Statistics-Importance and Need for Research Ethics		

	4	Selection of Topic for Research-Research schedules, Review of Literature and its Use in Designing a Research Work-		
	5	Mode of Literature Survey-Books and Monographs, Journals, Conference Proceedings, Abstracting and Indexing Journals, E-Journals/Books and CD-ROMS-Reports etc.		
	6	Thesis Writing		
	7	Computer Application in Scientific Research-www-Searching Scientific Articles		
	8	Statistical Data Base		
II	Scientific Word Processing with LaTeX and MS-Word		15	20
	9	Article, Thesis Report and Slides Making		
	10	Power Point Features, Slide Preparation		
	11	Statistical Programming with R: Simple Manipulations Using Numbers and Vectors-Objects & Their Attributes		
	12	Arrays and Matrices-Lists and Data Frames-Grouping, Loops and Conditions		
	13	User Defined Functions		
	14	Probability Distributions and Statistical Models in R		
III	Simulation		10	15
	15	Concepts and Advantages of Simulation		
	16	Event Type Simulation		
	17	Random Variable Generation-U(0,1), Exponential, Gamma and Normal Random Variables		
	18	Monte Carlo Integration		
	19	The MCMC Principle		
	20	Algorithms and its Variants, Bootstrap Methods		
IV	Computer Oriented Numerical Methods		8	15
	21	Algorithms for Solving Algebraic and Transcendental Equations		
	22	Numerical Integration		
	23	Matrix operations		
V	Open –Ended		12	
	Solve the problems from Module I to Module IV using software and understand how to check Plagiarism			

References

1. **Anderson, J., Durston, B.H., Pooole, M. (1970)** .Thesis and Assignment Writing. Wiley Eastern. Ltd., New Delhi.4
2. **Beveridge, B. (1979).** The Art of Scientific Investigation. W.E. Norton & Co., New York.
3. **Braun, J., Duncan, W. and Murdock, J. (2008).** A First Course in Statistical Programming with R. Cambridge University Press, London.
4. **Chambers, J. (2008).** Software for Data Analysis: Programming with R. Springer, New York.
5. **Crewley, M.J. (2007)** The R-.Book. John Wiley, New York.

6. **Dalgaard, P.(2008).** Introductory Statistics with R. Springer Science, New York.
7. **Kothari, C. (2005).** Research Methodology. New Age International. Publishers, New York.
8. **Lamport, L. (1999).** LATEX: A Document Preparation System. Addison, Wesley, 2nd edition, New York
9. **Panneerselvam. (2006).** Research Methodology. Prentice-Hall of India. Pvt.,New Delhi.
10. **Robert, C.P. and Casella, G. (2004).** Monte Carlo Statistical Methods. Springer Science, New York.
11. **Venkataraman, M.K. (1998)** Numerical Methods in Science and Engineering. The National Publishing Company, Chennai.
12. **Garg, B.L., Karadia, R., Agarwal, F. and Agarwal, U.K., (2002).** An Introduction to Research Methodology, RBSA publishers.